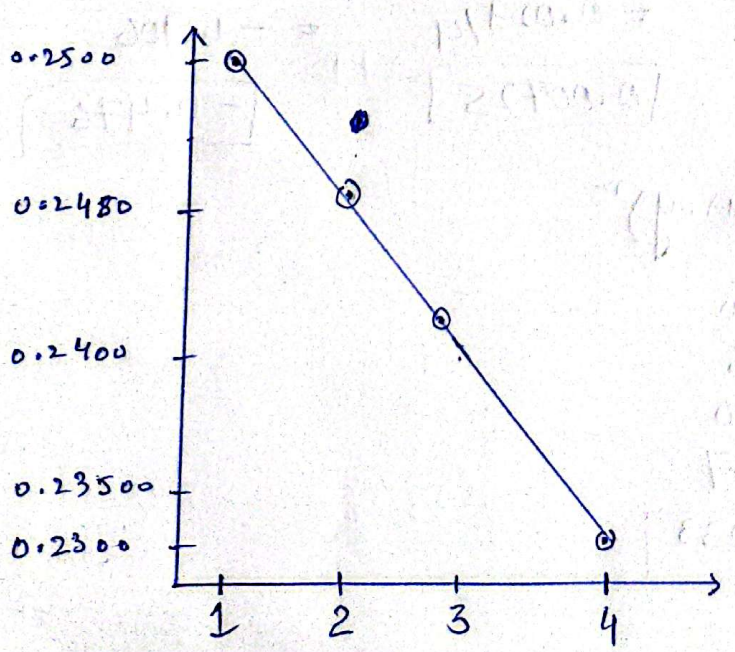
RESULTS

cost for 1-4 iterations.

No of iterations.	cost
1	0.2500
2	0.2480
3	0.2420
4	0.233



All mathematical practice is attached.

θ_0	θ_1
0	0

x_0	x_1	y
1	1	0
1	2	0
1	3	1
1	4	1

Learning rate

$\alpha = 0.01$

Iteration # 1

		$\theta^T x$	$h_\theta(x)$	y	$\frac{1}{m}\sum(h_\theta(x)-y)$	$\frac{1}{m}\sum(h_\theta(x)-y)x^i$
0	0	0	0.5	0	+0.5	0.5
0	0	0	0.5	0	+0.5	1
0	0	0	0.5	1	-0.5	-1.5
0	0	0	0.5	1	-0.5	-2
					0	-0.5

for cost $\frac{1}{m}\sum(h_\theta(x)-y)^2$

0.25
0.25
0.25
0.25

$1/4 = 0.25$

updating values of θ.

$\theta_0 = \theta_0 - \alpha(0) = 0 - 0.01(0)$ $\quad \theta_0 = 0$

$\theta_1 = \theta_1 - \alpha(-0.5) = 0 - 0.01(+0.5) = 0.005$

$\theta_1 = 0.005$

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Iteration 2 $\theta_0 = 0$ $\theta_1 = 0.005$

x_0	x_1	$\theta^T x$	$h\theta(x)(1/1+e^{-z})$	$\frac{1}{m}\sum(h\theta(x)-y)$	$\frac{1}{m}\sum(h\theta(x)-y)x^i$
1	1	0.005	0.501	0.501	0.501
1	2	0.01	0.502	0.502	1.004
1	3	0.015	0.503	-0.497	-1.491
1	4	0.02	0.505	-0.495	-1.98
				0.011/4 = **0.0027**	-1.972/4 = **-0.493**

For cost $\frac{1}{m}\sum(h\theta(x)-y)^2$

0.251
0.252
0.247
0.245

0.248

For updating "θ"

$\theta_0 = \theta_0 - 0.01(0.0027)$ **$\theta_0 = -0.000027$**

$\theta_1 = \theta_1 - 0.01(-0.493)$ **$\theta_1 = 0.0099$**

Iteration 3 $\theta_0 = -0.000027$ $\theta_1 = 0.0099$

x_0	x_1	$\theta^T x$	$h\theta(x)$	$\frac{1}{m}\sum(h\theta(x)-y)$	$\frac{1}{m}\sum(h\theta(x)-y)x^i$
1	1	0.00089	0.500	0.500	0.00044 0.500
1	2	0.0269	0.506	0.506	1.012
1	3	0.0539	0.513	-0.487	-1.461
1	4	0.0899	0.522	-0.478	-1.912
				0.0102	**-0.465**

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For cost $\frac{1}{m}\sum(h_\theta(x)-y)^2$

0.250
0.256
0.237
0.228

cost $\boxed{0.242}$

updating "θ"s.

$\theta_0 = -0.000027 - 0.01(0.0102)$ $\quad \boxed{\theta_0 = -0.00013}$

$\theta_1 = 0.009 - 0.01(0.465)$ $\quad \boxed{\theta_1 = 0.01365}$

Iteration 4

$\theta_0 = -0.00013 \quad \theta_1 = 0.01365$

x_0	x_1	$\theta^T x$	$h_\theta(x)$	$\frac{1}{m}\sum(h_\theta(x)-y)$	$\frac{1}{m}(h_\theta(x)-y)x^i$
1	1	0.0135	0.500	0.500	0.500
1	2	0.0271	0.506	0.506	1.012
1	3	0.0408	0.510	-0.49	-1.47
1	4	0.0544	0.513	-0.487	-1.948
				= 0.029/4	= -1.906
				$\boxed{0.00725}$	$\boxed{-0.476}$

for cost $\frac{1}{m}\sum(h_\theta(x)-y)^2$

0.250
0.256
0.2140
0.2137

$= \boxed{0.233}$